Road Extraction with InstaGeo

A Literature Review

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Abstract

Road extraction from remote sensing imagery is fundamental for urban planning, disaster management and environmental monitoring. This field remains challenging due to occlusions, class label imbalance, and the need for topological connectivity. Most studies and reviews in this field concentrate on high-resolution imagery due to the challenges associated with coarser grain imagery. As a result, this paper not only surveys deep learning methodologies with focus on state-of-the-art, and recent developments such as foundational models; but also focuses on medium resolution imagery and rural areas (as high-resolution imagery is less readily available on them). This study reveals the shift towards attention mechanisms and model efficiency, the under-utlilisation of medium resolution foundational models, the lack of benchmarking datasets for medium-resolution imagery, and the challenges that remain in the field. A comparative analysis of models evaluated on the DeepGlobe benchmark reveals the potential of large foundation models in road segmentation.

Keywords

Road Extraction, Semantic Segmentation, Remote Sensing

1 Introduction

Road extraction from remote sensing images remain challenging and relevant to researchers. The extracted network has practical applications in traffic monitoring, urban planning, disaster management routing, environmental monitoring, such as measuring ecological changes alongside road development.

Based on road features, road extraction can be divided into the following tasks: surface segmentation, edge detection, centreline extraction and their combination [46]. Road surface segmentation refers to classifying each pixel in the image as background or road, without differentiating between different road types. Most remote sensing road extraction papers focus on road surface extraction [35]. Road centreline extraction labels roads as a single-pixel wide line down the centre of the road and typically depend on surface segmentation results[40][13][52]. Road edge detection delineates the road with two lines, where most research focuses on automating driving and use data collected from vehicle-mounted sensors.

With the development of technology, remote sensing images are increasing in resolution and acquired in shorter cycles. The resultant datasets are more extensive, but also introduce more noise from non-road objects in the background and require more space-efficient processing methodologies. Furthermore, the inherent unbalanced nature of pixel labels (road versus background), occlusion from clouds, and shadows also add to the complexity. This review aims to (1) provide an overview of deep learning methods applied to remote sensing images for road extraction, (2) investigate the current state-of-the-art deep learning methodologies in the field, and (3) compare different data sources.

The scope of the review is limited to deep-learning methods applied to remote sensing images, with a particular focus on road surface segmentation methods applied on medium resolution images from 2023 to date and the relevance of geospatial foundation models. The remainder of the review is structured as follows. Section 2 discusses the datasets available. Section 3 discusses the relevant evaluation metrics. Section 4 outlines the deep learning methodologies and frameworks applied in the field. Section 5 discusses observations from the papers discussed. Section 6 concludes and identifies gaps in the research.

2 Data

2.1 High-Resolution Datasets

Road extraction research and surveys mostly centre around high-resolution data (<5m), because task complexity increases with a decrease in picture resolution. There are various datasets, as evident in Table 1. The two major benchmark datasets are as follows.

The Massachusetts dataset was the first large-scale public road detection dataset[42]. It covers urban, suburban and rural areas with varying terrains, landform features, and density of road networks. The dataset size makes it suitable for training medium-sized models.

The DeepGlobe [14] dataset is significantly larger than the Massachusetts dataset with most images obtained from Indonesia, Thailand, and India. It covers urban and rural settlements as well as various environments, such as sea shores and tropical forest. The diversity of images helps create more robust models.

The main disadvantage of high-resolution images is that they have relatively low revisit periods, expensive cost and small coverage. This limits their potential for large-scale road extraction and keeping up with rapid road changes[29]. Additionally, most datasets concentrate on urban areas.

2.2 Satellite

There are multiple large open-access datasets obtained from Earth Observation (EO) satellite missions. For example, some of the more impactful missions include Landsat, and Sentinel [67]. Additionally, The Satellite Needs Working Group created Harmonised Landsat and Sentinel-2 (HLS) to create a dataset with higher temporal resolution and is harmonious with Landsat's over 40 years of records. These datasets are often unlabelled and facilitate the unsupervised training of large foundational models.

Of the EO missions, Sentinel-2 currently is the state-of-the-art among global-coverage, open-data policy EO missions for optical imagery with its 10m resolution for RGB and NIR bands and 5-day

Table 1: Datasets for road extraction from remote sensing images

Dataset	Size	Train	Test	Val	Resolution	From	Time
Massachusetts[43]	1500 × 1500	1108	49	14	1	Massachusetts	2013
AerialKITTI-Bavaria [44]	512 × 512	360	100	–	–	AerialKITTI and Bavaria	2015
CNDS [13]	1600 × 1600	180	30	14	1.2	Google Earth	2017
CITY-OSM [30]	2611 × 2453	660	165	–	–	–	2017
RTDS [4]	4096 × 4096	25 cities	15 cities	–	0.6	Google Map	2018
DeepGlobe [14]	1024 × 1024	6226	1101	243	0.5	Thailand, Indonesia, India	2018
SpaceNet v3 [15]	1300 × 1300	2213	567	–	0.3	Paris, Las Vegas, Shanghai, Khartoum	2018
Conghua [59]	6000 × 6000	37	10	–	0.2	–	2019
RNBD [36]	–	14 regions	6 regions	1 region	0.21	Google Earth	2019
WorldView-4 [34]	512 × 512	6736	1012	–	0.31	–	2019
CityScale [23]	2048 × 2048	144	27	9	1	OSM	2020
Gaofen-2 [63]	512 × 512	36,000	4000	–	0.8	Fujian, Hainan	2020
ShaoShan [11]	1589 × 1131	29	20	–	0.5	Shaoshan	2020
GE-Road[49]	800 × 800	12,000	7000	1000	0.3–0.6	–	2020
CHN6-CUG [69]	512 × 512	3608	903	–	0.5	Google Earth	2021
LRSNY [12]	1000 × 1000	716	432	220	0.5	New York	2021
Icurb [60]	1000 × 1000	29,000	10,000	1000	0.152	NYC OpenData	2021
Sentinel-2 Road [31]	256 × 256	4507	563	564	10m	Sentinel-2	2025
WHU-RUR+[57]	1024 × 1024	18103	17995	-	0.3-0.8	-	2025
African Road Surface[37]	-	-	-	-	100-1000	OSM	2026

revisit rate. However, the main challenge for road extraction is that roads often appear thin or barely visible with subpixel road width.

2.3 Super Resolution

Super-resolution (SR) models are a deep learning-based approaches aimed at enhancing the spatial resolution of optical satellite images. Researchers sometimes use SR in tandem with road extraction algorithms to address the poor resolution of satellite imagery. Aybar et al. [3] categorised different SR approaches as SR with panchromatic reference, SR with multi-spectral reference and SR only with Low-resolution input. The last category is the most applicable in to road segmentation as it aims to enhance 10-meter resolutions as 5m or or less [3]. Aybar et al. [3] also provided their medium-sized model, SEN2SR, as a python package that includes upscaling resolution across all 10m and 20m bands of Sentinel-2.

2.4 OpenStreetMap (OSM)

OSM is a community-driven platform with global vector data. Particularly, its labelled road data can be aligned with satellite data at a large scale to facilitate road extraction at sub-pixel visibility. Although densely populated areas are relatively accurate and complete, rural areas annotations are often missing, outdated, or imprecise.

2.5 Data Augmentation

Data augmentation can be a useful tool in improving generalisability and robustness of models with the current available resources. Due to the nature of remote sensing data, images are often rotated, rescaled or flipped. More advanced techniques have started to employ generative models for more realistic data diversification. For example, Wang et al. [56] employs Stable-Diffusion-2-Inpainting to only alter background information in road datasets.

3 Metrics

This section will define the following metrics: accuracy, precision, recall, F1 score, intersection over union (IoU), mean intersection over union (mIoU), average path length similarity (APLS), entropy-based connectivity metric (ECM) and custom connectivity (CC). Strengths and weaknesses of the metrics will be briefly discussed.

Road surface segmentation is often a binary classification problem making confusion matrix-based metrics appropriate. A confusion matrix is made up of four quadrants: True Positive (TP), which is the number of correctly predicted road pixels; True Negative (TN), which is the number of correctly predicted non-road pixels; False Positive (FP), which is the number of incorrectly predicted road pixels; and False Negative (FN), which is the number of incorrectly predicted non-road pixels.

Accuracy (Equation 1) is the proportion of correctly classified pixels relative to the total number of pixels in the sample. A higher value reflects a better ability in correctly predicting road pixels. This metric is the most misleading as road segmentation datasets are inherently imbalanced and the background labels often overpower the road labels, which gives a false sense of the model's ability.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

Precision (Equation 2) is the proportion of correctly classified road pixels relative to the total number of road pixels predicted by the model. A higher value reflects better predictive ability for road pixels. It focuses on penalising the model for labeling background

pixels as road pixels.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall is the proportion of correctly classified road pixels relative to the total number of true road pixels. A higher value reflects a better ability to capture road pixels completely. It penalises the model for labeling road pixels as background pixels.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

F1 Score evaluates both accuracy and completeness of the model with the harmonic mean between Precision and Recall (Equation 4). The higher the value, the closer the model prediction is to the ground truth.

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

IoU (Equation 5) is the intersection of predicted and true road pixels, relative to the union of the predicted and true road pixels. It penalises the model for false positives and false negatives. Thus, a higher value reflects model results to be closer to the actual road.

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{5}$$

mIoU (Equation 6) is the average between the IoU of road and background pixels, where higher values indicate better road extraction ability.

$$mIoU = \frac{1}{2}\left(\frac{TP}{TP + FP + FN} + \frac{FN}{FN + TP + TN}\right) \tag{6}$$

The confusion matrix-based metrics described above are the most commonly used in research[15, 35]. However, these metrics do not encourage road connectivity. A model that predicts fully connected roads with thicker-than-ground-truth labels may appear to have poorer performance compared to a model that predicts roads perfectly, with some disconnected patches. The roads predictions given by the latter model would be less useful for routing use cases.

APLS [15] evaluates the accuracy of the predicted graph network. Dijkstra's shortest path algorithm is used to derive optimal paths. In equation 7, L(a,b) refers to the optimal path length(L) between nodes a and b on the ground truth graph. The node a' and b' are the nodes in the predicted graph closest to a and b, respectively. Higher values in APLS reflect a predicted road network graph that is more similar to the ground truth.

$$\text{APLS} = 1 - \frac{1}{N}\sum_{i=1}^{N} \min\left(1, \frac{|L(a,b) - L(a',b')|}{L(a,b)}\right) \tag{7}$$

ECM [61] is used to measure disconnections in the detection of road edges. The metric is defined in equation 8, where N is the total number of ground truth instances; α_i is the completion of the ith ground truth instance; C^i is the connectivity of the ith ground truth instance; M_i is the number of predicted road edges; and p_j is the jth predicted instance's dominance, which is the ratio of the connected edge, relative to the summed length of the predicted edges assigned to the ground truth edge. Larger values indicate more connectivity.

$$ECM = \sum_{i=1}^{N} \alpha_i e^{-C_i}, \quad C_i = \sum_{j=1}^{M_i} -p_j \log(p_j) \tag{8}$$

CC assesses the connectivity of road pixels in road segmentation predictions. It is defined as the total number of connected road pixels, N_c, divided by the total number of road pixels in the segmented region, N_t. Larger values indicate more connectivity of the extracted roads.

$$\text{CC} = \frac{N_c}{N_t} \tag{9}$$

4 Methodology

Traditional road segmentation techniques require manual feature extraction and offer limited scalability. As a result, deep learning techniques have dominated the field in recent years due to their automatic feature engineering and state-of-the-art performance, especially in larger datasets.

Various deep-learning methods have been implemented for the extraction of road data from remote sensing data. This section will briefly outline the development of deep learning techniques related to road segmentation.

4.1 Patch-based CNN

Patch-based CNNs first require images to be processed into patches, which are put into CNNs for feature extraction and road identification. Then, the patches are combined to output the predicted road network. Deep-learning, with patch-based CNN, was first applied to road extraction by Mnih and Hilton [43] in 2010. Though deep learning road segmentation started with CNNs, several other techniques have been developed and more commonly explored in recent years.

4.2 Encoder-Decoder Networks

CNN-based methods demonstrated superior performance compared to traditional non-deep learning methods when provided with sufficient data. However, labour-intensive and thus limited nature of high quality road segmentation datasets meant that more data efficient architectures were needed in the field. This led to the prominence gained by encoder-decoder models. Encoders are usually pre-trained classification networks that efficiently extract high-dimensional image features that decoders use to reconstruct the road network as output.

The emergence of FCNs[38] replaced the fully connected layers of CNNs with convolutional layers. These networks laid the foundation for pixel-level classification due to their end-to-end predictions per pixel[41], and improved network performance and accelerated network learning. The decoder of FCNs uses binary interpolation filters to reconstruct the feature map to match the input size. However, when applied to road segmentation, they struggled to adapt to new data distributions, especially of rural areas, due to the asymmetrical encoder-decoder structure and lack of high-resolution feature maps[56].

Proposed in 2015 for medical imaging, U-Nets [50] expanded on FCN and the encoder-decoder model ideas by incorporating skip connections between corresponding encoder and decoders. This ensures details, like road edges, are not lost and the model achieves more precise localisation. The architecture consists of a symmetric downsampling and upsampling path that resembled a "U". This encoder-decoder architecture helped capture both local and global contexts for effective segmentation. Currently, U-Nets

are still considered state-of-the-art, as emerging road extraction research most commonly benchmark proposed models against it.

In the early days of deep learning, most deep learning models improved performance with deeper neural networks. However, this makes optimisation more opaque. The development of deep residual learning framework [22] mitigated the vanishing gradient problem in deep networks by using skip connections with residual blocks to propagate information into earlier layers. This subsequently eased the optimisation process, and enabled efficient deeper networks to be developed.

LinkNet [6] was devised to enhance model efficiency for semantic segmentation using residual learning. The model design was to perform feature learning without significantly increasing parameters by replacing common encoders like ResNet101 and VGG16 with ResNet18. It directly transfers the extracted features of the encoder to the decoder without pooling or stride convolutions, which reduces processing while maintaining model accuracy.

Zhou et al. integrated LinkNet with dilated convolution layers to create D-LinkNet and won the CVPR DeepGlobe 2018 Road Extraction Challenge [68]. The dilated convolution was stacked in cascade mode into its central layers, which achieves larger receptive fields without downsampling high-resolution feature maps. However, the inherent sparse connections of the dilated convolution may have caused information loss [56].

ResUNet [66] combines the strengths of residual learning and U-Net specifically. Their proposed model outperformed U-Net in road segmentation while only having a fourth of its parameters.

4.3 Transformers

The wide adoption of Transformers [55] has garnered significant interest to attention mechanisms in road extraction due to their strong ability to acquire global context and leverage contextual cues through a self-attention mechanism. Road networks in remote sensing images are often relatively small in scale. Thus, vision transformers are integrated with neural network architecture, often "U" shaped CNNs, for road detection. This allows researchers to leverage the neural networks spatial analysis ability with the transformer's sequence data processing and long-distance dependency handling. The resulting effect includes better road connectivity, context-awareness and increased transferability.

Akhtarmanesh et al. [1] integrates a U-Net with attention mechanisms to leverage the global and local contexts as well as noise filtering, respectively. They integrated both hard and soft attention in the decoder to assist a U-Net for improved performance. For preprocessing on the DeepGlobe dataset, they patched images and masks, excluded images with very little to no road, and performed rotation-based augmentation. Hard attention was employed in preprocessing, for patching and background exclusion, while soft attention was employed in the model. When investigating model degradation, they categorised their predictions into four categories according to precision and recall. They attributed performance degradation with respect to over-masking and under-masking of ground truth labels, occlusion, unclear boundaries of rural roads affecting predicted width, falsely labeling road-like structures.

In an attempt to address occlusion, weak road features and complex background in road datasets, Ma et al. [41] integrates ResNet50 encoder, U-Net skip connections, strip convolution, and dilated convolution. ResNet50, lighter than ResNet101, balances performance requirements and model complexity. It also uses ImageNet preset weights [47] to accelerate model convergence and reduce misclassification of similar objects, a common problem, particularly for rural roads. Similar to D-LinkNet [68], dilated convolutions are placed at the centre. Strip convolution are more in line with the narrow, long shape of roads, compared to square convolutions that may include more unnecessary information when capturing a whole road. Though the model manages to capture roads similar to the background or obscured by vegetation, it still struggles to distinguish between dirt roads and dried riverbeds, as well as between roads and rivers.

For transferability, parameter efficiency and accuracy of rural road extraction, Wang et al. [56] developed TCNet using three attention components as follows. The clustered contextual Transformer module extended the shallow features. The clustered cross-attention module optimised global and local contextual information, specifically for rural roads. The convolution block attention module is integrated into encoder-decoder skip connection for efficient information transfer, and effective fusion between high-level and low-level features. To address the unbalanced nature of the data samples, they designed the Dice-BCE-Lovasz loss function. When TCNet's zero-shot transfer ability was compared to D-LinkNet and Deeplab v3+ on the Massachusetts dataset, their model demonstrated more connectivity and accuracy. Though, due to their focus on rural roads, their model has limited robustness on complex or occluded urban areas.

4.4 Foundation Models

Unsupervised learning does not rely on labelled data as they explore the structure and features of the input. Traditional unsupervised road extraction techniques such as edge detection, mathematical morphology etc. have been replaced with deep learning approaches for their ability to adapt to complex scenes[35].

Self-supervised learning, a type of unsupervised deep learning method, learns image features by formulating prediction tasks. Prevalent prediction task include contrastive learning and masked image modelling (MIM). Contrastive learning is where models try to identify images that are similar (created by the model through data augmentation) and different. MIM is done by the model covering up pixels for it to reconstruct and often applied using the Masked Auto-Encoder (MAE)[21]. Early foundational models often used contrastive learning. Nonetheless, MIM is replacing contrastive learning in remote sensing foundational models as it derives better feature representation and is more space efficient, which allows models to use more data easier[53]. The foundational models to be discussed are all based on MAE and include RingMo-SAM[62], Prithvi-EO-1[26], Prithvi-EO-2[54], and TerraMind[27].

RingMo-SAM is a foundational model trained on multimodal remote sensing data. It handles segmentation and classification in both optical and SAR data. As the name suggests, the model is based on the Segment Anything Model (SAM)[33], the first generalised segmentation foundation model with prompt engineering, and RingMo, a SSL foundation model framework with state-of-the-art performance on remote sensing downstream tasks. The

model was trained on datasets from multiple open-sources. The architecture includes a classification decoupling mask decoder for segmentation and classification. A prompt encoder is included to improve the segmentation performance of SAR. The model can identify object categories, and segment optical and SAR images. Limitations of the model include concatenation of objects in the remote sensing images and inferior SAR segmentation abilities compared to dedicated models.

Large foundational models provide generalised ability and aim to provide zero-shot or n-shot transfer of learning to more specific downstream tasks. Hence, task-specific fine-tuning are slowly being explored for road segmentation. This topic is of particular interest specifically due to the recent release of remote sensing foundation models.

Road-SAM [17] that extends SAM for road extraction in very high-resolution and large satellite remote sensing images. They proposed a modified SAM image encoder with mixed adapters to translate SAM from the natural image domain to the remote sensing domain, and an explicit visual prompt (EVP) designed for road extraction tasks through patch embedding and high frequency components to capture roads more accurately.

Other SAM-related road extraction are as follows. SAM_MLoRA [39] uses LoRA theory to fine-tune SAM for both building and road segmentation. Limitations of the model included disconnected roads due to background noise. The researchers attributed it to the lack of hard samples in datasets that intentionally introduce noise through small objects, such as roads obscured by trees and buildings. RN-SAM [32] proposes a framework to utilise OSM road network data for prompt-based fine-tuning and test time adaptation.

Following the release of SAM 2 [48] in 2024, methods such as SAM2MS [65] and SAM2-RoadNet [16] were developed. Both methods discard the prompt component of SAM to employ an adaptor-based fine-tuning for parameter efficiency. Additionally, SAM2MS employs subtraction blocks, and a training-free lossnet module for a multi-level supervision strategy. The subtraction blocks eliminate redundancy in during encoding and are cascaded to refine local feature representation. The model is described as robustness against noise interference, such as occlusion and shadows, though is sensitive to variation in resolution. SAM2-RoadNet integrates receptive field blocks for contextual perception enhancement and channel dimension alignment. Its weighted bidirectional feature pyramid network fuses the hierarchical features across different scales, and the topologically-aware soft-clDice loss reduces road fragmentation.

NASA and IBM collaborative developed Prithvi-EO-1 [26], an open sourced Geospatial Foundation Model (GFM) released with the trained model weights. It was trained on red, green, blue, NIR, SWIR 1 and SWIR 2 bands of the HLS 30m resolution remote sensing imagery of the US and access to the trained model weights allowed researchers to fine-tune the model for specfic downstream tasks by adding a task-specific decoder (also known as model head) to Prithvi's encoders [24]. Although the creators of Prithvi have demonstrated its superior performance compared to architectures such as U-Net in wildfire scar mapping and multi-temporal crop segmentation, there is a lack of application in the road extraction field.

This observation juxtaposes both SAM model's comparatively wide adoption in the field. This is likely due to the coarse 30m resolution of the HLS training data. Even at 10m resolution, road widths can reach sub-pixel size [2]. Fibaek et al. [18] noted that when applied to building density estimation, U-Nets captured finer details compared to Prithvi. Searching on Google Scholar using the keywords "Harmonized Landsat and Sentinel-2" AND "Road Extraction" yield very little results. In contrast, both SAMs were trained on data provided through human photographers with 3300x4950 pixels on average, making the model more attractive for capturing finer segmentation targets.

Thereafter, Prithvi-EO-2[54] was released with explicit transformer attention mechanism in both spatial and temporal dimensions, and focused on long terms trends and seasonal patterns. Despite the model's advancements, it is still trained on HLS data. Both versions of Prithvi are mostly applied to tasks that segment large patches of land, as suggested from the lack of smaller-scale object segmentation tasks given in their demo use cases [26, 54].

IBM also collaborated with the European Space Agency (ESA) to release TerraMind, the first any-to-any generative, multimodal GFM. This GFM is trained on the TerraMesh dataset[5], which is based on unlabelled remote sensing data that contain co-aligned radar and optical imagery from SSL4EO-S12 [5, 58] and MajorTOM-Core [19]. In contrast to the other two GFMs by IBM, this model's image source is from Sentinel-1 and -2 at 10m resolution. Due to its relatively new release, no road-related studies have been conducted.

4.5 Sentinel-based Research

Although road extraction efforts are mostly concentrated on high-resolution remote sensing images, there is an existing body of work that uses sentinel data at 10m resolution for road extraction.

For example, Oehmcke [45] conducted one of the earliest road detection attempts from Sentinel-2 imagery. With multimodal data as input, they leverage two U-Net-based models: one to detect roads in images with little occlusion and another to leverage temporal data to address occlusion and environmental variation. The first U-Net incorporates the higher resolution OSM vector data in the final upsampling layer. This allows the model to learn higher resolution road segmentation labels than the inputted Sentinel imagery. Instead of deconvolution in the upsampling layers, they implemented the pixel shuffle approach, used in creating super resolution images, to decrease checker board artifacts. The second model learns temporal dependencies by using two consecutive volumetric convolution layers to combine the spectral bands and time steps. The resultant IoU score is 0.217, 0.265, and 0.394 for small, medium and large roads, respectively with 0.053 - 0.111 units of improvement from the *vanilla* U-Net.

Similarly, Ayala et al. [2] also combines OSM annotations with Sentinel-2 imagery, though only the 10m RGB and NIR bands were used. Their approach includes appending a bicubic interpolation upsampling layer before the encoder U-Net to increase the input resolution (to 2.5), rather than increasing the output resolution. To balance the computational cost associated with the additional upsampling, they replaced the base encoder with ResNet34. The resulting F1 and IoU score of the *vanilla* U-Net and their proposal is 0.5345 and 0.3657 versus 0.8161 and 0.06894, respectively.

Also using OSM as ground truth, Gengec et al. [20] combines Sentinel-2 with GPS trajectory data, and explores early and late fusion techniques that occur at the beginning of the model and just before the output layer, respectively. They employed U-Net, ResUNet and D-LinkNet. Their experimental results reflected improved performance with the Sentinel-2 and GPS trajectory data fusion, particularly with late fusion. They found ResUNet to perform the best in their experiments and outperformed the benchmark set by Ayala et al. [2] by approximately 0.48 units.

The above studies [2, 20, 45] focused on urban areas, where OSM data was more accurate. However, rural roads are generally narrow, increasing the difficulty of road extraction using Sentinel RGB and NIR data. Jia et al. [29] uses a spatial relationship-informed super-resolution mapping network to obtain a finer scale classification map of rural roads (excluding highways and overpasses), despite the coarse satellite imagery. They proposed SRSNet, where a BaseNet extracts remote sensing image features, and a POINet extracts spatial features from rural settlement data. They leverage the co-existent relationship of rural roads with settlements and fuse the remote sensing image features with the spatial features using a feature enhancement model. An upsampling classification module is used for the super-resolving process and classification to achieve an IoU of 68.9%. The limitations of the study include small sample size, road dis-connectivity due to tree occlusion, misclassification near road edges that are unclear, and over-classification of roads leading to thicker roads extracted as an artifact of mixed image elements.

Building on [2], Jiangir et al. [28] combined super-resolution deep learning with bicubic interpolation to first improve the spatial resolution of Sentinel-2 data to 1.25m, then extract roads on the higher resolution images with the OneFormer model[25], a transform-based model. They reported 60.97% in road segmentation quality, a metric similar to IoU.

Shen et al. [51] proposes a dual-temporal farm road detection network model (DTBNet) using RGB Sentinel-2 data. Each branch of the model deals with a time step of an image. They used a two convolution encoder: standard square convolution to gather contextual information and dynamic snake convolution, similar to the strip encoder in the ASUNet [41] mentioned earlier, except it is deformable to curved local features to capture geometric and intrinsic features of the road. The snake-shaped convolution and dual-branch enabled more comprehensive feature extraction on the bi-temporal dataset. Attention Modules are used to fuse the generated features. Unlike the previous studies mentioned [2, 20, 28, 29, 45], no resolution adjustment was made to input images or output masks. The proposed model also generalises to high-resolution single-temporal datasets, such as Deep Globe. However, obstructing mountain shadows affected model performance, and tracks of harvesting machines were misclassified as roads.

4.6 InstaGeo

In light of the development of GFMs, InstaGeo [64] was developed as an open-source, end-to-end framework designed to unify the machine learning pipeline, particularly for GFMs. It aims to address the lack of preprocessing pipeline for satellite data extraction and impractical size of fine-tuned GFM. Their pipeline consists of three components, including data, model and application.

The data component automates the raw satellite image preprocessing to obtain model-ready data. It takes in geolocated points or polygons along with user defined spatial and temporal configurations to generate cloud-masked satellite data chips.

The model component transforms GFMs into task-specific models though vanilla fine-tuning GFMs or task-specific distillation for more compute-efficient models. Vanilla fine-tuning refers to fine-tuning a GFM where the resultant model retains the same architecture and parameter count regardless of the downstream task. The task-specific distillation is achieved through a student-teacher paradigm: a vanillar fine-tuned GFM model, with frozen weights, trains the student model to only retain the necessary encoders for specialised tasks. Through their distillation process, they observe an 8x smaller model size in the resultant model with less than 2 percentage point of performance loss. This lowers inference costs and the resultant CO2 emissions.

The application component is an interactive web-map application for users to deploy models. This includes selecting the area of interest on the web-map, running models based on the selected data, and exporting the model results as an analytical overview. This component promotes user-friendliness for non-technical users, especially for time-sensitive use cases such as disaster management.

Frameworks such as InstaGeo help enhance the practicality of GFM-based models in terms of user-friendliness and model efficiency.

5 Discussion

Deep learning in road extraction started with CNNs and is currently invested in transformers. As different techniques emerged, different components of the models were replaced or improved as researchers explore their potential. These included the emergence of residual learning, skip connections, dilated (atrous) convolution layers, attention mechanisms, and more. Throughout the body of research, some core technologies are identified as state-of-the-art and become the benchmark for evaluating model performance. Apart from U-Nets, various researchers have also benchmarked their model against ResUnet, LinkNet, Deeplab [7–10] and other models relevant to their specific research.

The majority of the research conducted in the field focuses on high-resolution optical remote sensing datasets in urban areas. Road datasets are inherently imbalanced with significantly more background pixels than road pixels. This challenge is exacerbated when roads are captured in medium resolution imagery, where road widths can be less than a pixel.

The studies using Sentinel data often leverage various spectral bands, time steps or additional information to supplement the poor resolution of the remote sensing images. In contrast to high-resolution datasets, which focus on urban areas, these studies are often in rural areas as the spectral properties of roads significantly differ to the surrounding environment, such as forests, farm land or grassland [51].

Other than performance, various researchers also emphasized model efficiency. This is especially important with the development of large vision foundation models, often with vision transformers

Table 2: Comparative performance of models on the DeepGlobe dataset. * DeepGlobe Wuhan

Model	Precision	Recall	F1	IoU	mIoU	APLS
D-LinkNet [68]	-	-	-	63.42	-	-
TCNet [56]	79.51	79.47	79.57	-	66.08	-
ROAD-SAM [17]	83.46	79.91	81.65	68.99	-	-
SAM_MLoRA* [39]	79.67	75.89	0.7773	0.6358	-	0.5342
SAM2MS[65]	79.50	81.32	81.31	-	64.24	-
SAM2-RoadNet [16]	80.12	81.56	80.79	-	64.84	68.72
ASUNet[41]	77.22	71.34	74.17	58..94	78.41	-
DTBNet[51]	78.8	81	79.9	67.8	-	-

as the backbone [5, 33, 48, 54], and fine-tuning them for various downstream tasks.

In table 2, a summary of model performance of all the papers that included DeepGlobe as their benchmarking is provided. These results show transformer-augmented and foundational model-based approaches generally outperformed earlier CNN-based encoder-decoder methods.

Specifically, foundation model adaptations based on SAM demonstrate notable improvements in segmentation accuracy. ROAD-SAM achieves the highest IoU (68.99) and F1 score (81.65) among reported methods, suggesting that large-scale pre-training on diverse datasets have the potential to enhance generalisation and feature representation.

Notably, DTBNet achieves a high IoU (67.8) and F1 (79.9) with Sentinel-2, without explicit resolution enhancement, as its training data. This may reflect the effectiveness of temporal feature fusion and dynamic snake convolution layers. Perhaps its dual-temporal design enables improved handling of seasonal variability, addressing a limitation in single-image approaches.

6 Conclusion

This paper provides a review of deep-learning-based road extraction from remote sensing imagery, from patch-CNNs to the emergence of large foundation models. The review focused on surveying recent medium resolution road extraction solutions and the potential of large foundation models, while highlighting state-of-the-art deep learning methodologies in the road extraction field.

Despite the advancements, challenges observed in road extraction include the inherent label imbalance of datasets, occlusion of adjacent objects, noise from shadows, misclassification of road-like objects, unclear boundary of rural roads, segmenting connected road networks, and imperfect ground truth labels.

The gaps identified in this review are as follows. (1) Most research and emerging architectures are applied to high-resolution remote sensing images. This trend is also evident in foundation model-based methods, where the SAM models are relatively extensively applied, but the Prithvi models and TerraMind do not have any road applications yet. (2) Lack of research conducted in both urban and rural road segmentation on Sentinel Data. Instead, most research is conducted where flora surrounds rural roads, likely due to the differing spectral quality that can be leveraged. (3) There is a lack of benchmarking datasets used in assessing rural segmentation, making it hard to compare different model performances. (4) There is a lack of multimodal and temporal datasets in high-resolution datasets that may help with seasonal and lighting variation.

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